FONTYS UNIVERSITY OF APPLIED SCIENCES
Dataspaces Project Team

Project Plan

Decentralized Dataspace Implementation

EDC & FIWARE Framework Deployment

Version	3.0
Date	June 17, 2026
Status	Updated
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1. Introduction

This project plan outlines the approach, timeline, and deliverables for the Decentralized Dataspace Implementation project carried out by the Dataspaces Project Team at Fontys University of Applied Sciences. The project runs from March 2026 through the end of June 2026.

1.1 Project Background

Organizations currently rely on centralized data silos that lack standardized, secure mechanisms for data exchange while preserving sovereignty. This project addresses the transition toward decentralized Dataspaces by deploying and evaluating two open-source connector frameworks: Eclipse Dataspace Components (EDC) and FIWARE Data Space Connector.

1.2 Design Challenge

Design a Decentralized Dataspace Infrastructure to enable Organizations and their Employees in a Collaborative Data Ecosystem to Securely Exchange and Govern Data in a Policy-Driven, Sovereignty-Preserving Environment.

1.3 Stakeholders

Stakeholder	Role	Involvement
Mark Klerkx	Direct Supervisor	Weekly guidance and feedback
Toon Teeuwen	Direct Supervisor	Weekly guidance and feedback
Bart van Gennip	Senior Stakeholder	Strategic oversight
DSSC	Reference Entity	Foundational reading and standards

2. Team Structure & Sub-Team Split

The team consists of four members. After completing Phase 1 (Technical Phase) collaboratively, the team splits into two dedicated sub-teams, each owning a framework end-to-end through use-case validation.

2.1 Sub-Team FIWARE (2-week sprint: Mar 23 – Apr 3)

Member	Primary Focus	Secondary Focus
Benjamin	FIWARE Deployment & Integration	Documentation, deployment guide
Juul	FIWARE IAM (DID, VC, Trust Anchors)	ODRL policies, combined project plan

2.2 Sub-Team EDC (5-week period: Mar 23 – Apr 25)

Member	Primary Focus	Secondary Focus
Georgi	EDC Deployment & Java Development	Testing & validation
Dimitar	EDC Use Case Design & Execution	Integration testing

After Sub-Team FIWARE completes their 2-week sprint (Apr 3), Benjamin and Juul join Sub-Team EDC to support the remaining EDC work, integration testing, and cross-framework validation.

3. Communication & Meetings

Day	Time	Purpose
Monday	09:00 – Full Day	On-campus work day: sprint planning, development, collaboration
Thursday	Afternoon	Progress check-in, blocker resolution, documentation sync

Primary communication channel: Microsoft Teams (Dataspaces Project Team).

All technical documentation is stored in the MS Teams shared files.

4. Project Phases & Timeline

The project is divided into four phases running from March 9, 2026 through June 26, 2026 (approximately 16 weeks). Phase 1 is complete. The core delivery work happens in Phases 2A (FIWARE) and 2B (EDC), which run in parallel with staggered durations. Phase 3 covers finalization and hand-off.

4.1 Phase 1: Technical Phase (Completed)

Duration	March 9 – March 20, 2026 (2 weeks)
Objective	Deploy functional EDC and FIWARE environments on Fontys Netlab, reachable via Fontys VPN.
Owner	All team members (collaborative)
Status	COMPLETED

Key Activities:

- Set up a local FIWARE Data Space Connector following the existing deployment guide.
- Set up Eclipse EDC environment using the official adopter documentation.
- Ensured both environments are reachable via Fontys VPN/Netlab.
- Evaluated feasibility of maintaining both frameworks.

Milestone: Both EDC and FIWARE connectors deployed, accessible via VPN, and framework evaluation decision made.

4.2 Phase 2A: Scenario Validation

Duration	March 23 – April 3, 2026 (2 weeks)
Objective	Validate end-to-end dataspace functionality on the both connectors across all defined use cases.
Sub-Team	Benjamin, Juul, Dimitar & Georgi
Priority	HIGH – Required for project success

FIWARE Use Cases to Validate:

UC-1: Provider-to-Consumer Data Transfer

- Configure a Provider connector and a Consumer connector within the FIWARE DSC.
- Register a data offering (e.g. a dataset or API endpoint) on the Provider side.
- Have the Consumer discover, negotiate, and successfully retrieve the data.
- Pass criterion: Consumer receives the correct, complete data from the Provider.

UC-2: Verifiable Credential Issuance & Verification

- Issue a Verifiable Credential (VC) to an organization/participant using the Trust Anchor / VC Issuer.
- Present the VC during a contract negotiation or data request.
- Verify the VC against the Trust Anchor before granting data access.
- Pass criterion: Access is granted only when a valid VC is presented; rejected when absent or invalid.

UC-3: ODRL Policy Enforcement

- Define an ODRL usage policy on a data offering (e.g. time-limited access, purpose restriction).
- Attempt a data transfer that complies with the policy – should succeed.
- Attempt a data transfer that violates the policy – should be blocked.
- Pass criterion: Policy engine correctly permits and denies access according to the ODRL rules.

UC-4: Marketplace / Catalogue Discovery

- Publish multiple data offerings to the FIWARE marketplace/catalogue.
- Browse, search, and filter offerings as a Consumer participant.
- Initiate a contract negotiation from the catalogue listing.
- Pass criterion: (A)Consumer is able to discover offerings and start a negotiation flow directly from the catalogue.

UC-5: Full Ecosystem End-to-End Test

- Run a complete scenario combining UC-F1 through UC-F4: a Consumer discovers data in the marketplace, authenticates with a VC, negotiates under an ODRL policy, and receives the data.
- Pass criterion: The entire flow completes without manual intervention; all governance checks pass.

Milestone: All five FIWARE use cases pass. Deployment guide and test results documented. Benjamin and Juul transitioned to support the EDC sub-team.

Duration	March 23 – April 25, 2026 (5 weeks)
Objective	Validate end-to-end dataspace functionality on the EDC connector across all defined use cases.
Sub-Team	Georgi, Dimitar (joined by Benjamin & Juul from Apr 4)
Priority	HIGH – Required for project success

4.4 Phase 3: Finalization Phase

Duration	April 28 – June 26, 2026 (9 weeks)
Objective	Replace dummy services with production-representative real-world services, finalize documentation, and prepare hand-off.
Owner	All team members
Priority	NICE-TO-HAVE – Strengthens the project but is not strictly required.

Key Activities:

- Replace placeholder/dummy containers with real-world representative services.
- Harden security configurations and IAM policies for realistic operation.
- Finalize and update all existing technical documentation.
- Prepare final demonstration and project hand-off materials.
- Conduct project retrospective and compile lessons learned.

Milestone: Production-representative dataspace environment with finalized documentation and project deliverables.

5. Timeline Overview

Wk	Dates	Phase	Key Milestone	Team
1	Mar 9 – 13	Technical	Environment setup started	All
2	Mar 16 – 20	Technical	Frameworks deployed, VPN confirmed	All
3	Mar 23 – 28	2A + 2B	Use-case work begins (parallel)	FIWARE: Ben+Juul | EDC: Georgi+Dimitar
4	Mar 30 – Apr 3	2A + 2B	FIWARE use cases complete (UC-F1–F5)	FIWARE wraps up; EDC continues
5	Apr 6 – 10	2B	EDC use cases (UC-E1–E3)	Full team on EDC
6	Apr 14 – 18	2B	EDC use cases (UC-E4–E5)	Full team on EDC
7	Apr 21 – 25	2B	EDC UC-E6 + integration testing	Full team on EDC
8–9	Apr 28 – May 9	Finalization	Real services replacing dummies	All
10–11	May 12 – May 23	Finalization	Documentation finalized	All
12–13	May 26 – Jun 6	Finalization	Final demo preparation	All
14–15	Jun 8 – Jun 19	Finalization	Hand-off materials	All
16	Jun 22 – 26	Finalization	Hand-off & retrospective	All

6. Use-Case Summary Matrix

The table below lists every use case that must be validated, its owning sub-team, target completion date, and acceptance criterion.

ID	Framework	Use Case	Due	Pass Criterion
UC-F1	FIWARE	Provider-to-Consumer Data Transfer	Apr 3	Consumer receives correct data
UC-F2	FIWARE	VC Issuance & Verification	Apr 3	Access granted/denied based on VC validity
UC-F3	FIWARE	ODRL Policy Enforcement	Apr 3	Policy engine permits/denies correctly
UC-F4	FIWARE	Marketplace / Catalogue Discovery	Apr 3	Consumer discovers and negotiates from catalogue
UC-F5	FIWARE	Full Ecosystem End-to-End	Apr 3	Complete flow without manual intervention
UC-E1	EDC	Provider-to-Consumer Data Transfer	Apr 25	Consumer receives correct data asset
UC-E2	EDC	VC Issuance & Verification	Apr 25	Only valid VC holders can negotiate
UC-E3	EDC	ODRL Policy Enforcement	Apr 25	Policy engine enforces all constraints
UC-E4	EDC	Federated Catalogue Discovery	Apr 25	Cross-provider discovery and negotiation
UC-E5	EDC	Full Ecosystem End-to-End	Apr 25	Complete flow with governance at every step
UC-E6	EDC	Cross-Framework Interoperability Smoke Test	Apr 25	Documented outcome with technical findings

7. Deliverables

#	Deliverable	Phase	Priority
1	Functional EDC environment on Netlab	Phase 1	**Required**
2	Functional FIWARE environment on Netlab	Phase 1	**Required**
3	Framework comparison and evaluation report	Phase 1	**Required**
4	Use-case test report on FIware (UC-F1–F5)	Phase 2A	**Required**
5	Use-case test report on EDC(UC-E1–E6)	Phase 2B	**Required**
6	Updated technical documentation (FIWARE + EDC)	Phase 1–3	**Required**
7	Combined project plan (IAM + Marketplace)	Phase 2	**Required**
8	Cross-framework interoperability findings	Phase 2B	**Required**
9	Production-representative environment	Phase 3	*Nice-to-have*
10	Final project hand-off package	Phase 3	**Required**

8. Risks & Mitigations

Risk	Likelihood	Mitigation
One framework proves infeasible to deploy	Medium	Plan already accounts for dropping to a single framework
IAM/VC integration more complex than expected	High	Juul leads IAM early in Phase 2A; leverage FIWARE Trust Anchor docs
Netlab/VPN infrastructure issues	Low	Coordinate with Fontys IT early; test access in week 1
Scope creep from real-world services	Medium	Phase 3 is explicitly nice-to-have; protect Phase 2 scope
FIWARE 2-week sprint proves too tight	Medium	Ben and Juul have prior FIWARE deployment experience; scope is validation, not greenfield
Cross-framework interop not protocol-compatible	High	UC-E6 is a documented smoke test; failure is an acceptable documented outcome

9. Actual Outcomes

- Eclipse MVD successfully deployed locally using KinD Kubernetes.
- Contract negotiation flow validated.
- DID resolution limitations identified in Netlab.
- EduCloud deployment achieved through Kubernetes and Traefik ingress.
- Dataspace Validation Dashboard publicly accessible.